VIKASH, VIKASH

CFD | Multiphase Flow | Fluid Dynamics | Reactive Flow | Heat Transfer | ANSYS Fluent | Star-CCM+



Address: Gothenburg, Sweden
Phone: +46 72 931 16 34
Email: vikasche.engg@gmail.com
LinkedIn: linkedin.com/in/vikash-vashisth-phd

SUMMARY

CFD specialist with a PhD in Chemical Engineering and more than four years of post-PhD experience in advanced CFD, multiphysics modelling and industrially relevant simulation. Expertise includes fluid mechanics, multiphase flow, turbulence modelling, reactive flow, multi-species transport and heat and mass transfer. Experienced in developing high-fidelity, mesh-independent CFD models and executing large-scale simulations in High-Performance Computing environments. My work covers the complete CFD cycle, including defining engineering questions, developing numerical methodologies, model setup, meshing, convergence assessment, post-processing, experimental validation and technical reporting. Experienced in translating complex simulation results into clear engineering insights, process improvements, design recommendations and quantified performance outcomes.

CORE COMPETENCIES

CFD / CAE Physics: Fluid Mechanics, Multiphase Flow, Multiphysics Modelling, Turbulence Modelling, Reactive Flow, Combustion-Adjacent Chemistry, Heat & Mass Transfer, Multi-Species Gas Chemistry, LES/RANS, Volume of Fluid (VOF), FEM, FVM, Plasma Modelling, Rotating Machine CFD, Electromagnetic Simulation

Numerical Methods: Mesh Generation, Mesh Refinement, Mesh Independence, Solver Convergence, Verification & Validation, Transient Simulation, High-Performance Computing (HPC) & Parallel Computing

Software & Scripting: ANSYS Fluent, STAR-CCM+, ANSYS Workbench, NX CAD, ANSYS HFSS, ANSYS PyFluent, Tecplot, UDF/C Scripting, Python, MATLAB, Jupyter Notebook, Linux, Technical Report Writing

PROFESSIONAL EXPERIENCE

Researcher / Postdoctoral Fellow – Industrial CFD & Multiphysics **Dec 2022 – present**
Chalmers University of Technology

Key Project 1: High-Temperature Multiphysics and Reactive Flow CFD

- Developed and validated multiphysics CFD models for plasma reactor, coupling thermo-fluid-chemical dynamics.
- Implemented multi-species chemistry methodology.
- Developed numerical methodology for intrinsic-quenching and forced jet-quenching configurations.
- Built mesh-independent, high-resolution models.
- Validated CFD predictions against experimental data.
- Quantified design impact: up to 70% reactor efficiency improvement against intrinsic quenched system.
- Achieved 10% NO yield gain and 6–8% energy-cost reduction against intrinsic quenched system through jet quenching.
- Delivered technical simulation reports and engineering design recommendations.
- Developing two-way electromagnetic-CFD workflows, one-way is operational.

Key Project 2: Multiphase Flow CFD – Fluid Dynamics in Turbulent Flows

- Developed and experimentally validated transient 3D multiphase CFD models, funded by the Swedish Research Council.
- Implemented the Volume of Fluid (VOF) method to resolve bubble and droplet breakup under turbulent flow conditions.
- Executed large-scale HPC simulations using approx. 2 million core hours to resolve sub-millisecond dynamics.
- Produced interface-resolved, mesh-independent simulations using adaptive mesh refinement.
- Validated incompressible multiphase-flow predictions against high-speed experimental measurements.
- Achieved micrometer-scale resolution of interfacial features beyond the spatial-resolution capabilities of the experimental setup.
- Quantified stochastic breakup behaviour, interface evolution, and satellite bubble and droplet formation.

Research Associate – Industrial Multiphase CFD **Jan 2022 – Apr 2022**
IIT Guwahati & Reckitt Benckiser, India

- Developed transient incompressible flow 3D CFD models of non-Newtonian shear-thinning rivulet flow.
- Implemented the Volume of Fluid (VOF) method to capture the falling liquid on inclined surface.
- Coupled interface transport with non-Newtonian rheology to predict free-surface flow behaviour.
- Quantified flow behavior and flow-regime transitions.
- Established mesh-independent solutions through systematic grid-refinement studies.
- Validated CFD predictions against experimental measurements.
- Delivered simulation-based process optimization recommendations.
- Delivered technical simulation report and findings, supporting industrial product formulation decisions.

Doctoral Researcher (PhD) – Rotating Machinery CFD, Hydrodynamics **Dec 2015 – Apr 2021**
IIT Roorkee, India

- Developed and validated transient sliding-mesh RANS and LES 3D CFD models for rotor-stator systems.
- Designed and evaluated a novel rotor-stator configuration using transient CFD simulations.
- Established mesh-independent solutions using systematic grid-refinement and targeted local-refinement strategies.
- Modelled incompressible flow using a pressure-based solver.
- Quantified hydrodynamic performance, pumping efficiency, energy dissipation, and flow behaviour across different rotor-stator geometries.
- Validated the predicted power number against benchmark data, achieving agreement within 10%.
- Developed stator geometry demonstrated measurable improvement vs baseline design.
- Modeled tracer and species transport to estimate mixing time for different fluids.
- Supervised master's students on CFD sub-projects, acting as a technical resource within the research group.

Assistant Manager – Manufacturing Engineering **Jul 2013 – Dec 2015**
Relaxo Pvt. Ltd., India

- Delivered approx. INR 10M in cost savings in FY 2013–14 via Lean Six Sigma process improvement, OEE analysis, and targeted automation of injection-molding cycle time.

EDUCATION

PhD – Chemical Engineering **2015 – 2021**
Indian Institute of Technology Roorkee, India

M.Tech – Chemical Engineering **2011 – 2013**
Indian Institute of Technology Roorkee, India

ADDITIONAL INFORMATION

- Languages: English – Fluent; Swedish – Intermediate, actively learning
- Technical reports delivered to NitroCapt AB (industrial partner) and Reckitt Benckiser (IIT Guwahati collaboration)
- Residential Fellowship – Institute of Advanced Studies, Loughborough University, UK
- National Fellowship (PhD & MTech) – Ministry of Education, India
- Driving License – Swedish Driving License (B)